Restaurant Inventory & Profit

Cactus Kitchen

100 pts

Unit 3 — Excel Spreadsheets · Variation Project Option B

/ 100 points · 40 tasks

How to use this instruction sheet: Download the starter file from the Project Grader, open it in Microsoft Excel 365, and keep this sheet beside you. Work through each task in order. Each task tells you what to do, why it matters, where to find it, and the correct formula syntax — but you must enter the formula yourself. The starter file contains clean data only — no instructions are embedded in the file itself.

E1 — Manage Worksheets & Workbooks

18 points (8 tasks)

2

Task 1 — Add a new worksheet to the workbook and rename it "Monthly Sales"

WHY THIS MATTERS

Organizing a workbook with multiple named sheets is a professional standard. Each sheet serves a distinct purpose and is clearly labeled.

WHERE TO FIND IT

Right-click any sheet tab at the bottom of Excel → Insert → Worksheet. Then double-click the new tab and type "Monthly Sales". Press Enter.

HOW YOU'LL KNOW IT WORKED

A new sheet tab labeled "Monthly Sales" appears at the bottom of the workbook.

2

Task 2 — Freeze row 3 so menu item headers stay visible while scrolling

WHY THIS MATTERS

With 20 menu items and many columns, freezing keeps column labels visible as you scroll down.

WHERE TO FIND IT

Click cell A4. View → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

A freeze line appears below row 3. Scroll down — column headers remain visible.

2

Task 3 — Navigate to the named range "MenuPrices" using the Name Box

WHY THIS MATTERS

Named ranges let you select important data ranges instantly.

WHERE TO FIND IT

Click the Name Box → type MenuPrices → press Enter.

HOW YOU'LL KNOW IT WORKED

The Price column cells for all menu items are selected.

2

Task 4 — Set the print area and set page orientation to Landscape

WHY THIS MATTERS

The inventory sheet is wide. Landscape orientation fits all columns on a single printed page.

WHERE TO FIND IT

Select the inventory data range. Page Layout → Print Area → Set Print Area. Then Page Layout → Orientation → Landscape.

HOW YOU'LL KNOW IT WORKED

Print Preview shows all columns fitting on one wide page.

2

Task 5 — Hide the internal cost columns; then unhide them for the manager review

WHY THIS MATTERS

Hiding cost columns protects margin information from non-management staff. Unhiding demonstrates reversibility.

WHERE TO FIND IT

Select Cost column → right-click → Hide. Then right-click adjacent column headers → Unhide.

HOW YOU'LL KNOW IT WORKED

The Cost column disappears, then reappears after unhiding.

2

Task 6 — Apply the "Ion" workbook theme

WHY THIS MATTERS

The Ion theme uses warm colors that complement the Southwestern restaurant brand of Cactus Kitchen.

WHERE TO FIND IT

Page Layout → Themes → find and click Ion.

HOW YOU'LL KNOW IT WORKED

The workbook updates to the Ion theme fonts and color palette.

2

Task 7 — Set workbook properties: Title = "Cactus Kitchen Inventory 2026", Author = your name

WHY THIS MATTERS

Inventory workbooks must be clearly identified for records management and sharing with vendors or accountants.

WHERE TO FIND IT

File → Info → Properties panel → fill in Title and Author.

HOW YOU'LL KNOW IT WORKED

File → Info shows the correct Title and Author.

4

Task 8 — Save the workbook as PDF for the owner and as CSV for the POS system

WHY THIS MATTERS

The owner needs a formatted PDF for records. The POS system needs a plain CSV — two formats for two audiences.

WHERE TO FIND IT

File → Save As → PDF for the owner. Then File → Save As → CSV for the POS copy.

HOW YOU'LL KNOW IT WORKED

Three files exist: the original .xlsx, a .pdf, and a .csv.

E2 — Manage Data Cells & Ranges

18 points (8 tasks)

2

Task 9 — Use AutoFill to extend "Item-01" and "Item-02" through "Item-20"

WHY THIS MATTERS

AutoFill recognizes the Item-01, Item-02 pattern and continues it automatically through Item-20.

WHERE TO FIND IT

Starter has "Item-01" in A4 and "Item-02" in A5. Select both. Drag fill handle down through row 23.

HOW YOU'LL KNOW IT WORKED

Column A shows Item-01 through Item-20 in all 20 menu rows.

2

Task 10 — Insert two new columns between Cost and Price for "Supplier" and "Reorder Level"

WHY THIS MATTERS

Supplier and reorder information belongs between cost and price in the inventory structure.

WHERE TO FIND IT

Right-click the Price column header → Insert (twice). Label the two new columns Supplier and Reorder Level.

HOW YOU'LL KNOW IT WORKED

Two new labeled columns appear between Cost and Price.

2

Task 11 — Format Cost and Price columns as Currency; format Margin column as Percentage

WHY THIS MATTERS

Currency shows dollar amounts clearly. Percentage converts 0.71 to 71% — the margin format managers expect.

WHERE TO FIND IT

Select Cost and Price → Home → $ Currency. Select Margin column → Home → % Percentage.

HOW YOU'LL KNOW IT WORKED

Cost and Price show $3.20 style. Margin shows 71.0% style.

2

Task 12 — Merge and center the title across the top row

WHY THIS MATTERS

A merged title spanning all columns makes the inventory workbook self-explanatory when shared.

WHERE TO FIND IT

Select row 1 across all columns. Home → Merge & Center. Type the title.

HOW YOU'LL KNOW IT WORKED

Row 1 is one wide cell with the title centered.

2

Task 13 — Wrap text in the Item Description column so full descriptions are visible

WHY THIS MATTERS

Without wrapping, long descriptions are cut off making the inventory incomplete as a reference.

WHERE TO FIND IT

Select the Item Description column. Home → Alignment → Wrap Text button.

HOW YOU'LL KNOW IT WORKED

Description cells expand vertically to show full text on multiple lines.

3

Task 14 — Apply conditional formatting to highlight items with margin below 20% in red

WHY THIS MATTERS

Low-margin items are financial risks. Red highlighting immediately flags items needing repricing.

WHERE TO FIND IT

Select the Margin column. Home → Conditional Formatting → Highlight Cells Rules → Less Than → type 0.2 → Red Fill.

HOW YOU'LL KNOW IT WORKED

Any menu item with margin below 20% has a red background.

2

Task 15 — Use Format Painter to copy category header formatting across all category rows

WHY THIS MATTERS

Format Painter ensures consistent formatting across all category rows without manual reformatting.

WHERE TO FIND IT

Click a formatted category header. Home → Format Painter (double-click to lock). Click each other category header. Press Escape.

HOW YOU'LL KNOW IT WORKED

All category header rows have identical formatting.

3

Task 16 — Name the menu price range "MenuPrices" for use in formulas

WHY THIS MATTERS

Named range enables readable formulas like =SUM(MenuPrices) and =AVERAGE(MenuPrices).

WHERE TO FIND IT

Select the Price column cells (F4:F23 after inserting columns). Formulas → Define Name → type MenuPrices → OK.

HOW YOU'LL KNOW IT WORKED

The Name Box dropdown shows MenuPrices as an available named range.

E3 — Manage Tables & Table Data

14 points (6 tasks)

2

Task 17 — Convert the menu item data range to an Excel Table

WHY THIS MATTERS

An Excel Table makes it easy to filter by category, sort by margin, and add total rows.

WHERE TO FIND IT

Click any cell in the menu data. Insert → Table → confirm range → OK.

HOW YOU'LL KNOW IT WORKED

Menu data has alternating row colors and dropdown filter arrows.

2

Task 18 — Apply an orange Medium table style to match Cactus Kitchen branding

WHY THIS MATTERS

Orange reflects the warm Southwestern aesthetic of Cactus Kitchen.

WHERE TO FIND IT

Click inside the table. Table Design → Table Styles gallery → find an orange Medium style → click it.

HOW YOU'LL KNOW IT WORKED

The table uses an orange color scheme.

2

Task 19 — Add a Total Row showing total revenue, total cost, and average margin

WHY THIS MATTERS

Total row summaries give the owner instant profit overview — total revenue, total cost, average margin.

WHERE TO FIND IT

Click inside the table. Table Design → Total Row checkbox. Set Price to SUM, Cost to SUM, Margin to Average.

HOW YOU'LL KNOW IT WORKED

A Total row shows sums for revenue and cost, and average for margin.

2

Task 20 — Filter the table to show only Dinner menu items

WHY THIS MATTERS

Filtering by category lets the chef review a specific meal period without distraction.

WHERE TO FIND IT

Click the Category column dropdown → uncheck Select All → check Dinner → OK.

HOW YOU'LL KNOW IT WORKED

Only the 6 Dinner items are visible.

2

Task 21 — Sort the full menu table by profit margin from highest to lowest

WHY THIS MATTERS

Sorting by margin descending identifies your most profitable items at the top.

WHERE TO FIND IT

Clear the filter first. Click the Margin column dropdown → Sort Largest to Smallest.

HOW YOU'LL KNOW IT WORKED

The highest-margin item appears at the top of the table.

4

Task 22 — Remove duplicate menu item entries

WHY THIS MATTERS

A duplicate menu item would inflate ingredient cost totals and distort revenue calculations.

WHERE TO FIND IT

Click inside the table. Table Design → Remove Duplicates → select all columns → OK.

HOW YOU'LL KNOW IT WORKED

Excel confirms 1 duplicate removed. The red duplicate row is gone.

E4 — Formulas & Functions

32 points (12 tasks)

2

Task 23 — Write =SUM(MenuPrices) to calculate total potential revenue if all items sold

WHY THIS MATTERS

Total menu revenue potential is a baseline for daily sales targets.

WHERE TO FIND IT

Click the yellow Total Revenue cell in the summary area.

FORMULA SYNTAX

```
=SUM(MenuPrices)
```

HOW YOU'LL KNOW IT WORKED

The cell shows the sum of all 20 menu item prices.

2

Task 24 — Write =AVERAGE(MenuPrices) for average menu item price

WHY THIS MATTERS

Average price shows where the menu is positioned compared to competitors.

WHERE TO FIND IT

Click the Average Price cell.

FORMULA SYNTAX

```
=AVERAGE(MenuPrices)
```

HOW YOU'LL KNOW IT WORKED

The cell shows the average price across all 20 items.

2

Task 25 — Write =MAX(MenuPrices) for highest priced item and =MIN(MenuPrices) for lowest

WHY THIS MATTERS

MAX and MIN show the price range — important for understanding customer expectations.

WHERE TO FIND IT

Click Highest Price → MAX. Click Lowest Price → MIN.

FORMULA SYNTAX

```
=MAX(MenuPrices) and =MIN(MenuPrices)
```

HOW YOU'LL KNOW IT WORKED

Highest shows $28.99 (Ribeye Fajitas). Lowest shows $3.49 (Agua de Jamaica).

2

Task 26 — Write =COUNTA(B4:B23) to count total number of menu items

WHY THIS MATTERS

COUNTA counts non-empty text cells — confirms all 20 items are entered.

WHERE TO FIND IT

Click the Total Menu Items cell.

FORMULA SYNTAX

```
=COUNTA(B4:B23)
```

HOW YOU'LL KNOW IT WORKED

The cell shows 20.

3

Task 27 — Write IF: if Margin >= 30% show "High Margin", else "Review"

WHY THIS MATTERS

IF automates margin classification. Items flagged Review need repricing or cost reduction.

WHERE TO FIND IT

Click the first Margin label cell in row 4. Copy down for all 20 items.

FORMULA SYNTAX

```
=IF((F4-E4)/F4>=0.3,"High Margin","Review")
```

HOW YOU'LL KNOW IT WORKED

Items with 30%+ margin show "High Margin". Others show "Review".

4

Task 28 — Write nested IF for reorder alert: Stock < 10 = "Order Now", Stock < 25 = "Low Stock", else "OK"

WHY THIS MATTERS

Nested IF evaluates stock levels. Clear reorder signals prevent running out of ingredients mid-service.

WHERE TO FIND IT

Click the first Reorder Alert cell. Copy down for all 20 items.

FORMULA SYNTAX

```
=IF(H4<10,"Order Now",IF(H4<25,"Low Stock","OK"))
```

HOW YOU'LL KNOW IT WORKED

Low stock items show the correct alert level based on their quantity.

3

Task 29 — Write COUNTIF to count how many items are in each menu category

WHY THIS MATTERS

Category counts tell the chef whether the menu is balanced across meal periods.

WHERE TO FIND IT

Click each category count cell in the summary area.

FORMULA SYNTAX

```
=COUNTIF(D4:D23,"Dinner")
```

HOW YOU'LL KNOW IT WORKED

Breakfast=3, Lunch=4, Dinner=6, Sides=2, Desserts=2, Beverages=2, Appetizers=1 (total=20).

3

Task 30 — Write SUMIF to total revenue by menu category

WHY THIS MATTERS

SUMIF totals prices for a specific category — shows which category generates the most revenue.

WHERE TO FIND IT

Click each category revenue cell.

FORMULA SYNTAX

```
=SUMIF(D4:D23,"Dinner",F4:F23)
```

HOW YOU'LL KNOW IT WORKED

Each category shows its total price sum. All combined equal SUM(MenuPrices).

3

Task 31 — Use MID to extract the position number from each item code (characters 6-7 of "Item-01")

WHY THIS MATTERS

MID extracts characters from the middle of a text string — creates a position number from the item code.

WHERE TO FIND IT

Click the first Cat Code cell.

FORMULA SYNTAX

```
=MID(A4,6,2)
```

HOW YOU'LL KNOW IT WORKED

"Item-01" returns "01". "Item-12" returns "12". All 20 items return their 2-digit position number.

2

Task 32 — Use TODAY() to display the inventory update date

WHY THIS MATTERS

TODAY() timestamps the workbook showing when data was last verified — important for food safety compliance.

WHERE TO FIND IT

Click the Inventory Date cell.

FORMULA SYNTAX

```
=TODAY()
```

HOW YOU'LL KNOW IT WORKED

The cell shows today's date and updates automatically each day.

4

Task 33 — Write XLOOKUP to pull the supplier name from the Suppliers reference sheet

WHY THIS MATTERS

XLOOKUP links inventory data to procurement information by looking up each item code in the Suppliers sheet.

WHERE TO FIND IT

Click the first Supplier cell (after inserting the Supplier column). Copy down for all 20 items.

FORMULA SYNTAX

```
=IFERROR(XLOOKUP(A4,Suppliers!$A$3:$A$15,Suppliers!$B$3:$B$15),"No Supplier")
```

HOW YOU'LL KNOW IT WORKED

Items with a supplier entry show the supplier name. 7 items show "No Supplier" — this is intentional.

3

Task 34 — Wrap the XLOOKUP in IFERROR to show "No Supplier" if item is not in the supplier list

WHY THIS MATTERS

IFERROR is already in the formula above. This task confirms you understand why — to handle the 7 items not in the supplier table gracefully.

WHERE TO FIND IT

The IFERROR is already in your formula. Verify that Item-11, 14, 15, 16, 17, 19, and 20 show "No Supplier".

FORMULA SYNTAX

```
=IFERROR(XLOOKUP(...),"No Supplier")
```

HOW YOU'LL KNOW IT WORKED

Exactly 7 items show "No Supplier". The other 13 show real supplier names.

E5 — Manage Charts

18 points (6 tasks)

3

Task 35 — Create a Bar chart showing revenue by menu category

WHY THIS MATTERS

Bar charts work well for category comparisons — category names fit better as horizontal labels.

WHERE TO FIND IT

Select category names and revenue totals from the summary area. Insert → Bar Chart → Clustered Bar.

HOW YOU'LL KNOW IT WORKED

A horizontal bar chart appears with one bar per menu category.

2

Task 36 — Move the revenue chart to its own sheet named "Revenue by Category"

WHY THIS MATTERS

A dedicated chart sheet displays at full size for owner reports and presentations.

WHERE TO FIND IT

Click chart. Chart Design → Move Chart → New sheet → type "Revenue by Category" → OK.

HOW YOU'LL KNOW IT WORKED

A "Revenue by Category" sheet tab appears with the chart at full size.

3

Task 37 — Change the profit margin chart type from Bar to Line to show trend

WHY THIS MATTERS

A Line chart shows the margin trend across items — revealing how quickly margins drop down the menu.

WHERE TO FIND IT

Click the margin chart. Chart Design → Change Chart Type → Line → OK.

HOW YOU'LL KNOW IT WORKED

The margin chart changes from bars to a connected line.

3

Task 38 — Add chart title "Menu Revenue by Category" and data labels

WHY THIS MATTERS

Data labels show exact revenue values on each bar so the owner can read precise numbers.

WHERE TO FIND IT

Click the revenue chart. Chart Design → Add Chart Element → Chart Title → type title. Then Add Chart Element → Data Labels → Outside End.

HOW YOU'LL KNOW IT WORKED

The chart has a title and each bar shows its exact revenue value.

3

Task 39 — Apply Quick Layout 2 and a dark chart style to the revenue chart

WHY THIS MATTERS

Quick Layout 2 positions the legend at the top. A dark style gives the chart a bold look for owner presentations.

WHERE TO FIND IT

Click chart. Chart Design → Quick Layout → Layout 2. Then Chart Design → Chart Styles → choose a dark style.

HOW YOU'LL KNOW IT WORKED

The chart uses Quick Layout 2 with a dark color scheme.

4

Task 40 — Add alt text to all charts in the workbook

WHY THIS MATTERS

Accessible documents are required for businesses working with government contracts or public services.

WHERE TO FIND IT

Right-click each chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED

Every chart has a descriptive alt text entry.